



PROCEDURE

Medication: Clozapine Coordinator Procedure – Clozapine Clinic

Scope (Staff):	North Metropolitan Health Service Mental Health (NMHS MH) Staff: Nursing/Medical
Scope (Area):	NMHS MH Community Mental Health Clinics (CMHC)

1. Aim

To provide a standardised and consistent approach to the management of clozapine patients mental and physical health by the clozapine coordinators in Community Mental Health Clozapine Clinics.

2. Background

Within NMHS MH CMHC the clinician best suited to undertake ongoing management of stable clozapine patients and engage other clinicians as appropriate is a clozapine coordinator. Once a patient is considered mentally and physically stable and on a maintenance dose of clozapine, the treating doctor may transfer clozapine management of the patient within the clinic to the clozapine coordinator, who will follow a protocol to monitor both the patient's mental and physical health, whilst also ensuring compliance with the ClopineCentral™ monitoring requirements.

3. Definitions

WBC: White blood cell

NC: Neutrophil count

GASS-C: Glasgow Antipsychotic Side effect Scale for Clozapine

BP: Blood pressure

BEN: Benign ethnic neutropenia

4. Risk

Not following these procedures can potentially lead to serious complications of clozapine.

5. Patient's Eligibility for Clozapine Clinic

For a patient to be eligible to be managed by the clozapine coordinator-led clinic the patient must meet the following criteria:

- Be on a stable dose of clozapine
- Be mentally stable
- Be physically stable
- Be assessed as suitable by the treating medical team.

6. Clozapine Monitoring Forms

At each clinic appointment, the clozapine coordinator and/or the doctor must use the following:

- Clozapine Monitoring [Form A](#)
- Clozapine Monitoring [Form B](#)
- GASS-C
- Constipation assessment

7. Management of Patients

7.1 Appointment schedule

Patients taking clozapine must be reviewed by a doctor or a qualified clinician every four weeks.

As a minimum, every sixth review must be conducted by a doctor.

The purpose of the 6-monthly medical appointment is to allow for regular oversight and assessment of clozapine monitoring, physical health monitoring and to allow for the patient to be seen for other medications/treatment that they may be receiving from the clinic that will require review.

7.2 Required at each appointment

Review blood test ensuring that WBC and NC are within the acceptable range (**WBC>3.5** **NC>2.0**).

This blood test will need to be retrieved and reviewed by the doctor.

At each appointment, the patient must be assessed using:

- [Clozapine Monitoring Form Part A](#)
- [GASS for Clozapine](#) side effects scale
- Constipation assessment ([Appendix 1](#)).

Assess the patient for:

- Signs of infection e.g. flu like or infective symptoms including fever, malaise and myalgia (potential signs/symptoms of clozapine-induced neutropenia/myocarditis).
- Signs of cardiac illness e.g. tachycardia, chest pain or shortness of breath, new onset fatigue (potential signs/symptoms of clozapine-induced cardiomyopathy/myocarditis).

Complete monthly physical health assessment:

- BP
- Heart rate
- Temperature (°C)
- Weight
- Waist circumference
- Ask about side effects using [GASS-C](#)
- Discuss bowel habits ([Appendix 1](#)).

Review and document:

- Mental health assessment
- Medication adherence
- Initiation of new medications in the previous month
- Caffeine consumption
- Cigarette smoking habits
- Constipation management including the use of laxatives

The following requires the coordinator to refer for immediate medical review:

- Blood tests in amber (WBC 3.00-3.5 or NC 1.5-2.0) or red (WBC<3.00 or NC<1.5) range (exception if patient has an override for BEN)
- Patient shows any signs or symptoms of infection/flu like illness or cardiac symptoms
- Patient has experienced a seizure
- Significant deterioration in mental state, coordinator has concerns about mental state
- Evidence of therapy interruption/non-adherence (>48 hours), blood testing (>72hours) or not attending appointments/unable to be contacted
- Pregnancy.

The following requires discussion with the doctor

(Immediate review may be necessary after details are discussed)

- Patient is experiencing new or troublesome side effects e.g. constipation
- Substantial change in caffeine consumption
- Change in smoking habits
- Abnormal physical health results
- Planning pregnancy or breastfeeding
- Patient/carer/support services or GP requests review.

7.3 Physical health monitoring

To be performed when required and as per monitoring protocol on Clozapine Monitoring Form B

Due to the potential adverse effects of clozapine ongoing physical health monitoring and management of adverse effects is essential. Results of ongoing physical health monitoring must be assessed by the treating doctor and where intervention is required referral to the patient's GP should be done. If the patient has no GP, the clozapine coordinator or the GP liaison officer will assist in helping the patient and case manager to organise a GP appointment for review.

A copy of the physical health monitoring forms must be filed in the patient's notes. If the patient's GP organises blood tests, then a copy of results needs to be obtained for the progress notes.

8. Clozapine Prescription

Clozapine prescriptions are generated by the patient's treating doctor. These can either be done at each appointment after the review or as a prescription with up to 5 repeats with a phone authority. In case of the latter, the patient's nominated pharmacy will hold the repeats, and not dispense to the patient until the required monthly blood test is viewed by the pharmacist and has been entered into ClopineCentral™.

9. Pharmacist

The Clozapine Clinic Pharmacist may be contacted for assistance Monday to Wednesday on Ph: (08) 6159 6677.

The State-wide Psychotropic Drug Information Service may be also be consulted for assistance:
Telephone: (08) 6159 6400. Monday – Friday 08:30-16:30.
Email: DrugInformation.Graylands@health.wa.gov.au

Related internal policies, procedures and guidelines
NMHS MH Medication: Clozapine Prescribing and Monitoring NMHS MH Medication: Clozapine Management by General Practitioners

References
Mental Health Charts and Clozapine Resources ClopineCentral™ Protocol

Useful resources (including related forms)
GASS for Clozapine Clozapine Monitoring Form Part A Clozapine Monitoring Form B The Royal Australian & New Zealand College of Psychiatrists



Policy Sponsor	Chief Pharmacist, NMHS MH				
Policy Contact	Chief Pharmacist, NMHS MH				
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Approved by:	NMHS MH Drugs and Therapeutics Committee			Date:	04/10/2022
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NSQHS Standards Applicable:	☒  Std 1: Clinical Governance ☒  Std 4: Medication Safety		☒  Std 5: Comprehensive Care ☒  Std 8: Recognising and Responding to Acute Deterioration		
National Standards for Mental Health Services	☒ Std 1: Rights and Responsibilities ☒ Std 2: Safety ☒ Std 3: Consumer and Carer Participation		☒ Std 8: Governance, leadership and management ☒ Std 10: Delivery of Care		
Chief Psychiatrist's Standards for Clinical Care:	Assessment, Care Planning, Consumer and Carer Involvement in Individual Care, Physical Health Care of Mental Health Consumers, Risk Assessment and Management, Transfer of Care				
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Appendix 1. Constipation Assessment

The normal frequency of bowel motions can vary between individuals from 3 times per day to 3 times per week.

Therefore constipation should not be diagnosed based on the frequency of bowel motions alone.

Constipation should be assessed using the following criteria in conjunction with the Bristol Stool chart.

Constipation can be confirmed when 2 or more of the following are present:

- two or fewer motions per week
- straining at more than 25% of defecations
- a feeling of incomplete evacuation at 25% or more defecations
- hard lumpy stools (1 or 2 on the Bristol Stool Chart)
- sensation of anorectal obstruction or blockage for at least 25% of defecations
- manual manoeuvres needed to facilitate at least 25% of defecations (e.g. digital evacuation, support of the pelvic floor).

It is important to review current medication therapy and ask about the use of over the counter medications, complementary medicines and laxatives to assist in identifying possible causes of constipation and determine appropriate management.

Other common issues that could be exacerbating or causing constipation should also be assessed including:

- review other medication that could be causing constipation (particularly anticholinergics)
- inadequate dietary fibre
- dehydration (e.g. not drinking enough water)
- inappropriate bowel habits (e.g. ignoring the urge to defecate)
- inadequate physical activity (e.g. from decreased mobility)
- change in environment (e.g. holiday, entry into a residential aged care facility)
- painful anorectal disorders (e.g. haemorrhoids, anal fissure)
- loss of muscle power caused by other medical conditions (e.g. chronic obstructive pulmonary disease).

Reference: [Therapeutic Guidelines > Gastrointestinal > Constipation > Functional gastrointestinal disorders > functional constipation](#).



Bristol Stool Chart

Type 1		Separate hard lumps, like nuts (hard to pass)
Type 2		Sausage-shaped but lumpy
Type 3		Like a sausage but with cracks on the surface
Type 4		Like a sausage or snake, smooth and soft
Type 5		Soft blobs with clear-cut edges
Type 6		Fluffy pieces with ragged edges, a mushy stool
Type 7		Watery, no solid pieces. Entirely Liquid

First published: Lewis SJ, Heaton KW (1997) Stool form scale as a useful guide to intestinal transit time. Scandinavian Journal of Gastroenterology 32: 920–4

Type 1–2 indicate constipation Medical team should review and provide treatment

Type 3–4 are ideal stools as they are easier to pass, and

Type 5–7 may indicate diarrhoea and urgency. – Review and if indicated there may be the need to cease/reduce laxatives or consider symptoms as a sign of overflow.